

Def. Let  $X, Y$  be Metric Spaces, and  $f_n: X \rightarrow Y$  be function defined for each  $n \in \mathbb{N}$ .  
For each  $x \in X$ , define  $f(x) \stackrel{\text{def}}{=} \lim_{n \rightarrow \infty} f_n(x)$  if the limit exists.

This  $f$  is called limit function of  $\{f_n\}$ .

Def. Suppose that a sequence of functions  $\{f_n\}$  on  $X$  into  $Y$  satisfies that:  
the limit function  $f$  is defined on  $X$  and

For all  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.  $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \epsilon, \forall x \in X$ .

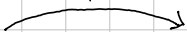
Then, we say that  $\{f_n\}$  converges uniformly on  $X$  to  $f$ . Denote  $f_n \rightrightarrows f$ .

Note:  $\forall x \in X, d_Y(f_n(x), f(x)) \leq \epsilon \Leftrightarrow \sup_{x \in X} d_Y(f_n(x), f(x)) \leq \epsilon$ . That is,

$$f_n \rightrightarrows f \text{ on } X \Leftrightarrow \lim_{n \rightarrow \infty} \sup_{x \in X} d_Y(f_n(x), f(x)) = 0.$$

Thm. Let  $\{f_n\}$  be functions defined on any Metric Space  $X$  into Complete Metric space  $Y$ . Then,

In General Metric Space



$\{f_n\}$  converges uniformly on  $X$  to  $f$  if and only if  $\forall \epsilon > 0, \exists N \in \mathbb{N}$  s.t.  $m, n \geq N \Rightarrow \sup_{x \in X} d_Y(f_n(x), f_m(x)) \leq \epsilon$

In Complete Metric space

Proof. If  $f_n \rightrightarrows f$ , then,

$$\sup_{x \in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x \in X} d_Y(f_n(x), f(x)) + \sup_{x \in X} d_Y(f(x), f_m(x)) \rightarrow 0$$

Give the result. (간단하게 보려면, Supremum의 성질과 triangle inequality가 활용되면 풀 수 있음.)

Conversely, since Cauchy sequence in the Complete Metric space converges, the limit function  $f$  exists.

Given  $\epsilon > 0$ , choose  $N \in \mathbb{N}$  such that the Cauchy condition holds.

For fixed  $n \geq N$ , for all  $x \in X$  and  $m \geq N$ ,  $d_Y(f_n(x), f_m(x)) \leq \epsilon$ .

So, letting  $m \rightarrow \infty$ , we obtain  $d_Y(f_n(x), f(x)) \leq \epsilon$ , for all  $x \in X$ . Thus proved.  $\square$

Fail Try.

$$\lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x)) = \sup_{x \in X} d_Y(f(x), f_n(x))$$

Gives the result. (More specifically, let  $n \in \mathbb{N}$  fixed. Observe that

$$\left| \sup_{x \in X} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f(x), f_n(x)) \right| \xrightarrow{\text{d}_Y \text{ continuous}}$$



$$\sup_{x \in X} d_Y(f(x), f_n(x)) = \sup_{x \in X} d_Y(\lim_{m \rightarrow \infty} f_m(x), f_n(x)) = \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) = \lim_{m \rightarrow \infty} \sup_{x \in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x \in X} \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - \sup_{x \in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x \in X} \left| \lim_{m \rightarrow \infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \rightarrow 0$$

Gives (4).

# Uniform Convergence and Continuity.

Thm. Suppose that  $f_n, f: X \rightarrow Y$  are functions where  $X$  is Metric,  $Y$  is complete Metric sp.

Let  $x \in X$  be a limit point of  $X$ .

If for each  $n \in \mathbb{N}$ , the limit  $\lim_{t \rightarrow x} f_n(t)$  exists and  $f_n \rightarrow f$  uniformly on  $X$ ,

then  $\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t)$

Proof. For each  $k \in \mathbb{N}$ , take  $A_k = \lim_{t \rightarrow x} f_k(t)$ . And, since the uniform convergence, given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$m, n \geq N \Rightarrow d_Y(f_n(t), f_m(t)) \leq \epsilon \text{ for all } t \in X.$$

And, for such  $m, n \geq N$ , let  $x_0 \in X$  such that

$$d_Y(A_m, f_m(x_0)) < \epsilon \text{ and } d_Y(A_n, f_n(x_0)) < \epsilon$$

(for some  $\delta, \delta > 0$ ,  $t \in B_\delta(x) \Rightarrow d_Y(f_k(t), A_k) < \epsilon$ , by def. of  $A_k$ )

Now,

$$d_Y(A_m, A_n) \leq d_Y(A_m, f_m(x_0)) + d_Y(f_m(x_0), f_n(x_0)) + d_Y(f_n(x_0), A_n) \leq 3\epsilon,$$

Therefore  $\{A_k\}$  is a Cauchy sequence in  $Y$ , so it is convergent.

Let  $A = \lim_{n \rightarrow \infty} A_n$ . Then,

$$d_Y(f(t), A) \leq \underbrace{d_Y(f(t), f_n(t))}_{f_n \rightarrow f \text{ uniformly}} + \underbrace{d_Y(f_n(t), A_n)}_{f_n(t) \rightarrow A_n \text{ as } t \rightarrow x} + \underbrace{d_Y(A_n, A)}_{A_n \rightarrow A} \rightarrow 0$$

Rigorously, given  $\epsilon > 0$ , let  $N_1, N_2 \in \mathbb{N}$  s.t.

$$n \geq N_1 \Rightarrow d_Y(A_n, A) < \epsilon \text{ and } n \geq N_2 \Rightarrow \forall t \in X, d_Y(f_n(t), f(t)) < \epsilon.$$

Now, take  $n \geq \max\{N_1, N_2\}$  and  $\delta > 0$  s.t.

$$t \in B_\delta(x) \Rightarrow d_Y(f_n(t), A_n) < \epsilon. \quad \blacksquare$$

Corollary. If  $\{f_n: X \rightarrow Y\}$  is a sequence of continuous maps on a Metric space  $X$  into Complete  $Y$ , and  $f_n \rightarrow f$ , then  $f$  is continuous.

## Uniform Convergence with Continuity

Thm. Suppose that  $f_n$  ( $n \in \mathbb{N}$ ) are functions on a Metric  $X$  into Complete Metric  $Y$ . Let  $x \in X$  be a limit point of  $X$ .

If for each  $n \in \mathbb{N}$ , the limit  $\lim_{t \rightarrow x} f_n(t)$  exists and  $f_n$  converges uniformly on  $X$ , then 
$$\lim_{t \rightarrow x} \lim_{n \rightarrow \infty} f_n(t) = \lim_{n \rightarrow \infty} \lim_{t \rightarrow x} f_n(t).$$

Proof. By the Uniformly Convergence, given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  s.t.

$$n, m \geq N \Rightarrow d_Y(f_m(t), f_n(t)) \leq \epsilon \text{ for any } t \in X.$$

Thm. Let  $K$  be a Compact space, and

$f_n: K \rightarrow \mathbb{R}$  be continuous functions which converges pointwise to a continuous function  $f$  on  $K$ .

If  $f_n \geq f_{n+1}$  for all  $n \in \mathbb{N}$ , then  $f_n \rightarrow f$  uniformly.

Proof. Put  $g_n = f_n - f$ . Then,  $g_n$  is continuous and  $g_n \geq 0$  pointwise, and  $g_n \geq g_{n+1}$ .

Given  $\epsilon > 0$ , let  $K_n = \{x \in K \mid g_n(x) \geq \epsilon\} = g_n^{-1}([\epsilon, \infty))$

Since  $g_n$  is continuous,  $K_n$  is a closed set in a Compact space, so it is compact.

And, since  $g_n \geq g_{n+1}$ ,  $K_n \supseteq K_{n+1}$  (if  $g_{n+1}(x) \geq \epsilon$ , then  $g_n(x) \geq g_{n+1}(x) \geq \epsilon$ )

Fix  $x \in X$ . Since  $g_n(x) \rightarrow 0$  as  $n \rightarrow \infty$ , for large enough  $n \in \mathbb{N}$ ,  $x \notin K_n$ .

That is,  $x \notin \bigcap K_n$ , so  $\bigcap K_n$  is empty. But, since each  $K_n$  is compact, thus  $K_N = \emptyset$  for some  $N \in \mathbb{N}$ . That is,

$$0 \leq g_N(x) < \epsilon \text{ for all } x \in K,$$

and since  $g_N \geq g_{N+1} \geq g_{N+2} \geq \dots$ , so  $\sup |g_n - 0| \rightarrow 0$ .  $\square$

Thm. Let  $X$  be a Metric space, and  $\mathcal{C}_b(X)$  be the all complex-valued continuous, bounded functions on  $X$ .

Define  $\|f\| = \sup_{x \in X} \|f(x)\|$ , then it is norm on  $\mathcal{C}_b(X)$ . This norm can derive a Metric on  $\mathcal{C}_b(X)$ .

## Uniform Convergence and Integration.

Thm. Let  $\alpha: [a, b] \rightarrow \mathbb{R}$  be monotone increasing and  $f_n \in \mathcal{B}(\alpha)$ ,  $n \in \mathbb{N}$ .

If  $f_n \rightarrow f$  on  $[a, b]$ , then  $f \in \mathcal{B}(\alpha)$  and

$$\lim_{n \rightarrow \infty} \int_a^b f_n d\alpha = \int_a^b \lim_{n \rightarrow \infty} f_n d\alpha = \int_a^b f d\alpha.$$

Proof. For each  $n \in \mathbb{N}$ , define  $E_n = \sup_{x \in [a, b]} |f_n(x) - f(x)|$ . Then, clearly

$$f_n - E_n \leq f \leq f_n + E_n \quad (\text{for all } t \in [a, b])$$

$$\text{Therefore, } \int_a^b f_n - E_n d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f_n + E_n d\alpha \quad (*)$$

$$\text{Hence } 0 \leq \int_a^b f d\alpha - \int_a^b f d\alpha \leq 2E_n \cdot [\alpha(b) - \alpha(a)] \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Thus  $f \in \mathcal{B}(\alpha)$ . The equality is obtained by (\*), using  $f \in \mathcal{B}(\alpha)$ .

## Uniform Convergence and Differentiation.

Thm. Let  $f_n: [a, b] \rightarrow \mathbb{R}$  be differentiable maps s.t. for some  $x_0 \in [a, b]$ ,  $\{f(x_0)\}$  converges.

If  $\{f_n\}$  converges uniformly on  $[a, b]$ , then  $\{f_n\}$  converges uniformly on  $[a, b]$  to  $f$  with

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x).$$

Proof. Given  $\varepsilon > 0$ , take  $N \in \mathbb{N}$  s.t.  $n, m \geq N$  implies

$$|f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} \text{ and } |f'_n(t) - f'_m(t)| \leq \frac{\varepsilon}{2(b-a)}, \quad \forall t \in [a, b].$$

By the Mean-Value Theorem for  $f_n - f_m$ , for all  $x, t \in [a, b]$ ,

$$|f_n(x) - f_m(x) - (f_n(t) - f_m(t))| \leq \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}. \quad (*)$$

Therefore, for any  $x, t \in [a, b]$  and  $n, m \geq N$ ,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

so that  $\{f_n\}$  converges uniformly, let  $f = \lim_{n \rightarrow \infty} f_n$ . For each  $x \in [a, b]$ , define on  $[a, b] \setminus \{x\}$

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x} \quad \text{and} \quad \phi(t) = \frac{f(t) - f(x)}{t - x}$$

(\*) gives  $\phi_n$  is Cauchy sequence, thus converges uniformly, to  $\phi$ . Finally,

the fact that  $\phi_n$  continuous gives the result.

Summary. Let  $f_n: X \rightarrow Y$  be functions,  $f_n \rightarrow f$ .

Continuity If  $f_n$  are continuous and  $f_n \rightarrow f$ , then  $f$  is continuous.

Differentiation If  $f_n$  are differentiable and  $f_n \rightarrow f'$  and  $f_n(x_0) \rightarrow f(x_0)$ , then  $f_n \rightarrow f$  and

$$\frac{d}{dx} \lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{d}{dx} f_n(x)$$

Integral If  $f_n \in \mathcal{B}(a)$  and  $f_n \rightarrow f$ , then  $f \in \mathcal{B}(a)$  and

$$\int_a^b \lim_{n \rightarrow \infty} f_n \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n \, dx$$

# Counterexamples

$$\begin{aligned} & \Gamma f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto x^n & f_n & \text{Conti,} \\ & \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1. \end{cases} & f & \text{not Conti} \end{aligned}$$

$$\begin{aligned} & \Gamma f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n}x & f_n & \text{converges uniformly} \\ & \overline{f_n}: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{1}{n}x & \overline{f_n} & \text{not converges uniformly} \end{aligned}$$

$$\begin{aligned} & \Gamma f_n: [0,2] \rightarrow \mathbb{R}; x \mapsto \frac{1-x}{1+x^n} & f_n & \text{differentiable, converges uniformly} \\ & \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1-x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } 1 < x \leq 2. \end{cases} & f & \text{not differentiable at } x=1 \end{aligned}$$

$$\text{In } [0,1], |f_n(x) - f(x)| = \left| (1-x) \cdot \left( \frac{1}{1+x^n} - 1 \right) \right| = \frac{|1-x| \cdot |x^n|}{|1+x^n|} \leq |1-x| \cdot |x^n| = (1-x) \cdot x^n.$$

$$\begin{aligned} & \text{Since } ((1-x) \cdot x^n)' = -x^n + (1-x)n \cdot x^{n-1} = nx^{n-1} - (n+1)x^n = x^{n-1}(n - (n+1)x) \\ & \text{has a maximum at } x = \frac{n}{n+1}, \quad \left( 1 - \frac{n}{n+1} \right) \cdot \left( \frac{n}{n+1} \right)^n = \frac{n^n}{(n+1)^{n+1}} \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

$$\text{So, } \sup_{[0,1]} |f_n(x) - f(x)| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\text{In } [1,2], |f_n(x) - f(x)| = \left| \frac{1-x}{1+x^n} \right| = \frac{x-1}{1+x^n} \leq \frac{x-1}{x^n} = \frac{1}{x^n} - \frac{1}{x^n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Now,  $f_n \rightarrow f$  converges uniformly.

$$\begin{aligned} & \Gamma f_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{1}{\sqrt{n}} \cdot \sin(nx). & f_n & \text{differentiable, converges uniformly,} \\ & f = 0, \text{ since } |\sin(mx)| \leq 1 & & \text{but} \\ & f'_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sqrt{n} \cdot \cos(nx) & f'_n & \text{not converges pointwise.} \end{aligned}$$

$$\Gamma \text{ Let } X_n = \{x_n | n \in \mathbb{N}\} = \mathbb{Q} \cap [0, 1].$$

$$f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 1 & \text{if } x \in X_n \\ 0 & \text{if } x \notin X_n. \end{cases} \quad f_n \text{ Riemann-integrable, since it has finite discords.}$$

$$f: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases} \quad f \text{ not locally integrable, it is Dirichlet.}$$

$$\Gamma f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} n & \text{if } 0 < x < \frac{1}{n} \\ 0 & \text{if } x=0 \text{ or } \frac{1}{n} \leq x \leq 1. \end{cases}$$

$$\lim_{n \rightarrow \infty} f_n \equiv 0, \text{ but } \int_0^1 f_n = 1. \quad \text{Cannot change limit under the integral.}$$

$$\Gamma f_n: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{x^2}{(1+x^2)^n} \quad f_n \text{ continuous, but the sum is not continuous.}$$

$$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n} = \begin{cases} 0 & \text{if } x=0 \\ 1+x^2 & \text{if } x \neq 0 \end{cases} \quad \left( x^2 \cdot \frac{1}{1 - \frac{1}{1+x^2}} \right)$$

$$\Gamma f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto nx \cdot (1-x^2)^n \quad f_n \text{ differentiable,}$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0$$

$$\int_0^1 x(1-x^2)^n dx = \int_1^0 -\frac{1}{2} \cdot t^n dt = \frac{1}{2} \cdot \frac{1}{n+1} t^{n+1} \Big|_0^1 = \frac{1}{2} \cdot \frac{1}{n+1}.$$

$$\text{So, } \int_0^1 f_n(x) dx = \frac{1}{2} \cdot \frac{n}{n+1} \rightarrow \frac{1}{2}, \text{ but } \int_0^1 f dx = 0.$$

$$\text{Note: replace } nx \text{ to } n^2 x, \text{ then } \int_0^1 f_n \rightarrow \infty.$$